Reg. No. | | | | | | | | | | | [17EC43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

SIGNALS AND SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

i. **Answer any FIVE full questions by choosing at least TWO questions from each part.**

ii. **Any missing data may be suitably assumed.**

PART - A

1. **a.** Give the detailed classification of signals with a suitable example for each. **05**

b. Given $x(t)$ and $y(t)$as shown in fig 1b, construct the following. **06**

i. $x(t+1)\,y(t-2)$

ii. $x(2t)\,y(-t-1)$

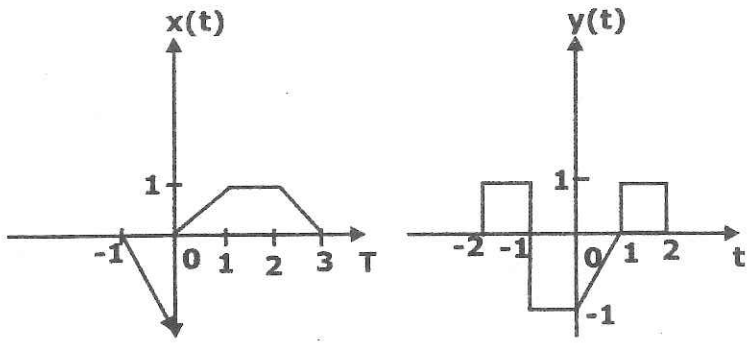

Fig.1b

c. For the following discrete signals, determine whether each signal is periodic or not. If it is periodic, find its fundamental period. **03**

i. $x[n] = 5\sin[2n] + 2\cos[0.2\pi n]$

ii. $x[n] = 3\sin[6\pi n] + 4\cos[\frac{3\pi n}{5}]$

2. **a.** Prove the condition for linearity and time invariance of a system. Hence derive an expression for output of an LTI system. **06**

b. Given the impulse response of a LTI system,$h[n] = (1/2)^n u[n-2]$, solve for its output for the following inputs. **08**

i. $x[n] = u[n] - u[n-6]$

ii. $x[n] = u[n+5] - u[n-6]$

3. **a.** Solve for the forced response of the system represented by the following difference equation. **08**

$y[n] - \left(\frac{4}{9}\right) y[n-2] = x[n-1]; \qquad x[n] = 2u[n]$

Also give its Direct form I and Direct form II block diagram representation.

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b. For the following impulse responses, show whether the corresponding system is **06**

i. memory less,

ii. causal, and

iii. stable. Justify your answers.

1) $h(t) = u(t+1) - u(t-1)$

2) $h[n] = 2^n u[-n]$

05

4. a. Identify the appropriate Fourier representation for each of the following signals.

i. $x[n] = (\alpha)^n u[n-4]$

ii. $x[n] = \sum_{k=-\infty}^{\infty}(\delta[n-22k] - 2\delta[n-2-22k])$

iii. $x(t) = 1 - \cos 2\pi t + \sin 3\pi t$

iv. $x(t) = e^{-t}\cos(2\pi t)u(t)$

b. For the given time domain signal, sketch Fourier domain amplitude and phase spectra. Ref fig. 4b. **09**

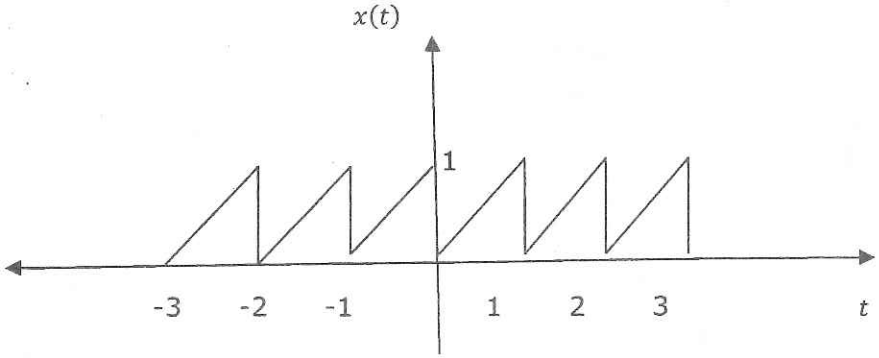

Fig.4b

PART-B

5. a. Use the convolution and time - differentiation properties to find the FT of **04**

$$y(t) = \frac{d}{dt}\{te^{-3t}u(t) * e^{-2t}u(t)\}$$

b. State and prove convolution and time shift properties of Fourier transform. **05**

c. Show that the Fourier transform of a rectangular pulse described by: **05**

$x(t) = 1;\ -T_0 \le t \le T_0$

$\quad = 0;\ \ |t| > T_0$

is a sinc function. Plot the magnitude and phase spectra.

6. a. The input $x(t) = e^{-5t}u(t)$, when applied to a system, results in an output $y(t) = e^{-t}u(t)$. Find the frequency response and impulse response of the system. **05**

b. Explain the process of sampling and concept of aliasing as applicable to continuous time signals. **05**

c. Find inverse Fourier transform of $X(\omega) = \frac{j\omega+12}{(j\omega)^2+5j\omega+6}$ **04**

7. a. Determine z-transform of the following signals. Identify the ROC in each case and locate the poles and zeroes. Comment on ROC versus nature of time domain discrete signal. **09**

i. $x[n] = \left(-\frac{1}{2}\right)^n u[-n] + 2\left(\frac{1}{4}\right)^n u[n]$

ii. $y[n] = \left(-\frac{1}{2}\right)^n u[n] + 2\left(\frac{1}{4}\right)^n u[n]$

iii. $w[n] = \left(-\frac{1}{2}\right)^n u[-n] + 2\left(\frac{1}{4}\right)^n u[-n]$

b. Determine the inverse z-transform of $X(z) = \frac{1}{1-1.5z^{-1}+0.5\,z^{-2}}$, ROC $|z| < 0.5$. Use partial fraction expansion method. **05**

8. a. Discuss in detail relevant transform domain relations applicable to analog amplitude modulation process. **07**

b. How do we characterize a communication filter system using Fourier concepts? Explain with examples. **07**